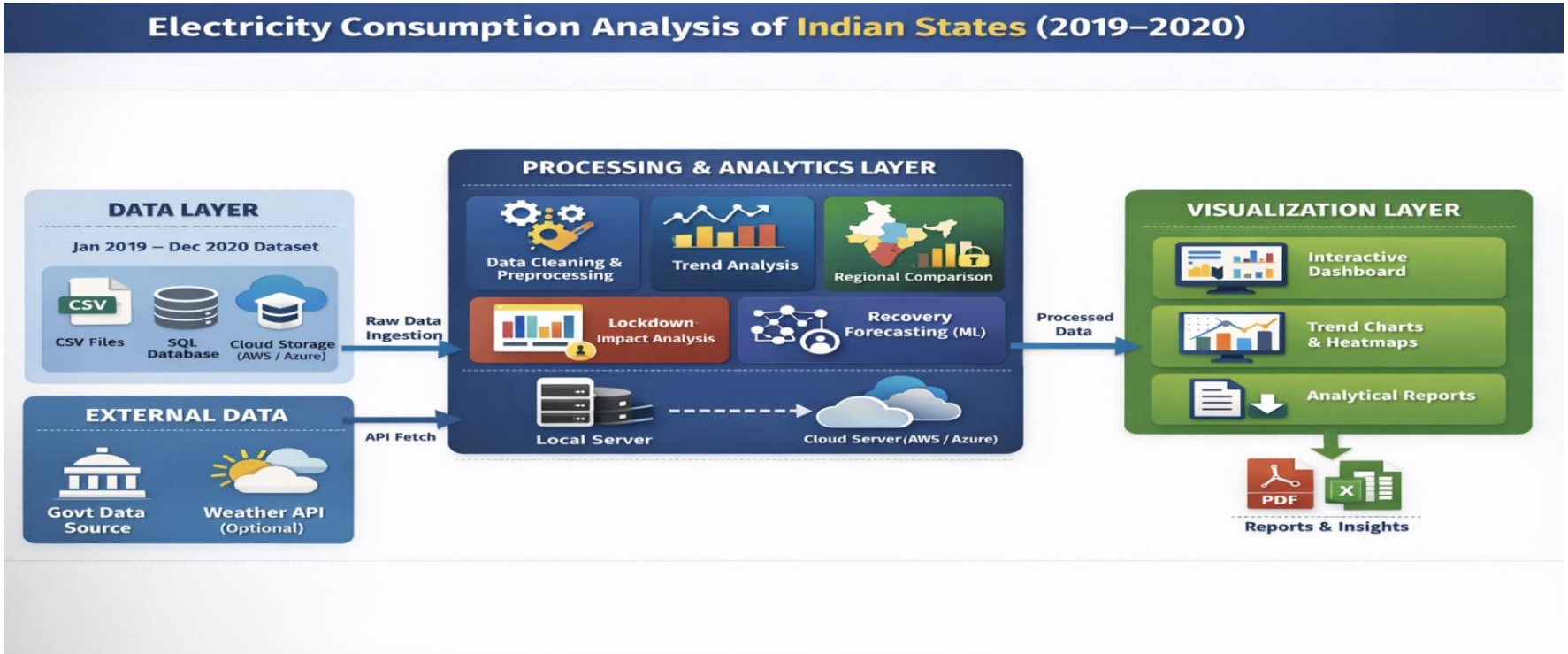
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 FEBRUARY
Team ID	
Project Name	**An Exploration of Electricity Consumption Patterns**
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information in table 1 and table 2



Guidelines – How the System Works

1) Data Collection (Data Layer)

- Collect state-wise electricity data (Jan 2019 – Dec 2020)
- Data sources: Government, CSV Files, SQL Database, Cloud storage (AWS / Azure)

CSV

Cloud storage

Output: Raw structured dataset

2) Data Cleaning & Preprocesing

- Remove duplicate entries
- Handle missing values
- Convert date column into time-series format
- Standardize state and region names
- Detect & treat outliers

Output: Cleaned & validated dataset

3) Processing & Analytics Layer

- Analyze trends & **regional differences**
- Calculate lockdown recovery rates
- Optional: Demand forecasting (ML)

Output: Processed analytical insights

4) Visualization Layer

- Generate **interactive dashboard** (Power BI / Tableau)
  - Line Charts
  - Bar Charts
  - Heatmaps

5) Infrastructure

- Local deployment
- Cloud deployment (AWS EC2 / Azure VM)

Output: Robust and scalable system

6) Performance & Scalability

- Optimize database operations
- Implement 3-tier architecture
- Allow future data integration

Output: Efficient handling of large datases

End Result of the System

Demand Shifts	Lockdown Impact	Regional Variation	Recovery Pattern	Recovery Pattern	Forecasting Capability

Clear & actionable insights for electricity consumption analysis of **indian states (2019–2020).**

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard for visualization	Power BI / Tableau / Streamlit
2.	Application Logic-1	Data Cleaning & Preprocessing	Python (Pandas, NumPy)
3.	Application Logic-2	Trend & Regional Analysis	Python (Matplotlib, Seaborn)
4.	Application Logic-3	Lockdown Impact & Recovery Analysis	Python (Statistical Analysis)
5.	Database	Structured monthly electricity data	MySQL / PostgreSQL
6.	Cloud Database	Remote database storage	AWS RDS
7.	File Storage	Raw dataset storage	Local Filesystem / AWS S3
8.	External API-1	Weather Data (optional correlation)	OpenWeather API
9.	Machine Learning Model	Electricity Demand Forecasting	Scikit-learn / ARIMA
10.	Infrastructure (Server / Cloud)	Deployment Environment	Local Server / AWS EC2

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Data analytics and visualization frameworks	Python, Pandas, NumPy, Matplotlib
2.	Security Implementations	Secure login, encrypted data storage	SHA-256, HTTPS, IAM Controls
3.	Scalable Architecture	Secure login, encrypted data storage	Cloud Deployment (AWS)
4.	Availability	Cloud hosting ensures high uptime	AWS EC2 + RDS
5.	Performance	Optimized queries, caching, efficient data handling	SQL Indexing, Pandas Optimization

References: